

Patrick Pastorelli

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MSc in Computer Science (University of Trento) specializing in embedded low-power systems and hardware–software co-design. Author of two IEEE publications in robotics and intermittent computing, including a first-author paper in RA-L 2025. Experienced competitive programmer and SWERC 2022/2023 regional contestant.

Research Interests

My research interests focus on intermittent computing and energy harvesting for battery-free IoT systems. I have research experience the design of intermittent systems and ultra-low-power communication, including wake-up radios, as well as modular firmware for resource-constrained devices. I also have research experience in robotic path planning and autonomous navigation.

Experience

IEEE RA-L

Invited Reviewer

2026

Peer review of multiple Nonholonomic Motion and Path Planning manuscripts

University of Trento

Internship in PCB Design with Altium Designer

2025

Full PCB design cycle using Altium Designer schematic capture and 2-layer layout design and optimization.

University of Trento

Teaching Assistant in Computer Science

2025

Weekly tutoring sessions for undergraduate students, providing support in C++ debugging, memory management, and algorithmic logic during laboratory sessions and assisted official examinations.

University of Trento

Research Collaborator

Trento, Italy

2024–2025

Collaborated with Prof. Luigi Palopoli and Prof. Davide Brunelli on two main research tracks:

- Robotics: Developed a novel polyline smoothing algorithm for robot path planning, leading to a publication in IEEE RA-L.
- Embedded Systems: HW-SW co-design of an intermittent computing scheduler for the CapDYN smart battery system, presented at IEEE Sensors 2025.

University of Trento and CovisionLab (Bozen, Italy)

Image compression pipeline in CUDA with CovisionLab

2024

Development and analysis of a high-speed image compression pipeline to accelerate data ingestion of AI models using CUDA. The project also included an evaluation of CUDA Optical Flow.

Publications

2026: [Energy-Latency Characterization of an Integrated Wake-Up Radio SoC](#)

P. Pastorelli, D. Brunelli, and M. Nardello.

ENSsys 2026.

To Appear

- 2025:** [Every Microjoule Counts: Zero-Failure Task Execution in Batteryless Sensors](#)
M. Nardello, M. Doglioni, S. Dagnino, **P. Pastorelli**, and D. Brunelli.
IEEE Sensors 2025.
Currently invited by IEEE Sensors Journal for a Special Issue on Selected Papers from the IEEE Sensors 2025 Conference
- 2025:** [Fast Shortest Path Polyline Smoothing With G1 Continuity and Bounded Curvature](#)
P. Pastorelli, S. Dagnino, E. Saccon, M. Frego and L. Palopoli.
IEEE Robotics and Automation Letters, Volume: 10, Issue: 4.
Also presented at IRIM 2024 and IROS 2025

Technical Skills

Languages: C++23, Rust, C, Python, Bash, CUDA, Haskell

Embedded: STM32, ESP32, TI MSP430, FreeRTOS, ROS2, HAL/LL Drivers, PCB Design (Altium)

Wireless: UWB, LoRa, IEEE 802.15.4, Sub-Ghz, WURs

Tools: Git, Linux, Docker, LaTeX, MATLAB, Makefile

Education

University of Trento

Trento, Italy

Master's Degree in Computer Science, 110/110 cum laude (GPA 4.0)

2023–2025

- Focus: Development and experimental validation of firmware and drivers for battery less, intermittent computing systems, power profiling using digital oscilloscopes and logic analyzers, robotics systems.
- Thesis: Exploring a Flooding-Based Wake-Up Radio Routing Protocol on a New Ultra-Low Power SoC for the IoT.

University of Genova

Genova, Italy

Bachelor's Degree in Computer Science, 110/110 cum laude (GPA 4.0)

2020–2023

- Focus: Team working, Operating Systems, Computer Architecture, Software Engineering.
- SWERC: Member of the first team at SWERC 2022/2023.

RUI Foundation

Genova, Italy

JUMP, Job-University Matching Project

2020–2023

Interdisciplinary that focuses on the development of transferable skills across three main areas: soft skills, modern themes, and interdisciplinary courses.

Others

Languages: Italian (Native), English (C1 level, FCE 182), Chinese (A1)

Interests: Former debater, also studied chess for 12 years, participating in multiple tournaments organized by FIDE such as the international chess festival of Imperia. Currently studying mandarin Chinese with the goal of obtaining an HSK certification in the future.